

Discussion on K-Medoids Clustering

- ❑ K-Medoids Clustering: Find representative objects (medoids) in clusters
- ❑ PAM (Partitioning Around Medoids: Kaufmann & Rousseeuw 1987)
 - ❑ Starts from an initial set of medoids, and
 - ❑ Iteratively replaces one of the medoids by one of the non-medoids if it improves the total sum of the squared errors (SSE) of the resulting clustering
 - ❑ PAM works effectively for small data sets but does not scale well for large data sets (due to the computational complexity)
 - ❑ Computational complexity: PAM: $O(K(n - K)^2)$ (quite expensive!)
- ❑ Efficiency improvements on PAM
 - ❑ CLARA (Kaufmann & Rousseeuw, 1990):
 - ❑ PAM on samples; $O(Ks^2 + K(n - K))$, s is the sample size
 - ❑ CLARANS (Ng & Han, 1994): Randomized re-sampling, ensuring efficiency + quality

K-Medians: Handling Outliers by Computing Medians

- ❑ Medians are less sensitive to outliers than means
 - ❑ Think of the median salary vs. mean salary of a large firm when adding a few top executives!
- ❑ K-Medians: Instead of taking the mean value of the object in a cluster as a reference point, medians are used (L1-norm as the distance measure)
- ❑ The criterion function for the K-Medians algorithm

$$S = \sum_{k=1}^K \sum_{x_i \in C_k} |x_{ij} - med_{kj}|$$

- ❑ The K-Medians clustering algorithm:
 - ❑ Select K points as the initial representative objects (i.e., as initial K medians)
 - ❑ Repeat
 - ❑ Assign every point to its nearest median
 - ❑ Re-compute the median using the median of each individual feature
 - ❑ Until convergence criterion is satisfied